



BASE II Clearing Visa Clearing Exchange (VCX)

Client Implementation Guide

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Visa Supplemental Requirements

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Chapter 1

Introduction to VCX

This provides specific system and onboarding information for participating acquirers, issuers, and processors to implement Visa Clearing Exchange (VCX).

VCX is a software replacement for Edit Package, Release 4.0. VCX is provided by Visa to VisaNet endpoints for processing transaction files sent to, and received from, VisaNet. VCX acts as the interface between the client processing center's internal payment processing system and VisaNet. It prepares the endpoint's processing center's outgoing clearing files for transmission to Visa. VCX also receives, and reports on, incoming clearing files from Visa, preparing it for the processing center's host system.

Overview of VCX

This gives an overview of Visa Clearing Exchange (VCX).

Primary Features

VCX includes these primary features:

- Windows Operating System
- Linux Operating System
- Client-server architecture

- Secure framework
- Browser-based user interface
- Flexible reporting options
- Batch procedure customization via run control options.

Note: VCX Release 1.0 will not be backwards compatible with any previous versions of Edit Package.

Changes to Manage

Clients who choose to implement VCX should be prepared for these changes:

- Linux and Windows installation procedures
- Distribution of installation software and updates
- Hardware and software requirements
- System administration flows
- User configuration and setup
- Report viewing, archiving, and printing
- Run control management

VCX Support

VCX support is available for clients should they encounter any issues. For example, when the Action portion of an error message description in Visa Clearing Exchange (VCX) says to contact Visa for support, you should contact your Visa representative. Before contacting Visa Technical Support, you should have this user and system information available for the support technician.

- Your name, phone number, and email address.
- Basic information about your processing center, such as its name, BASE II CIB, location, supervisor, phone number, and email address.
- A full description of the problem, including the exact steps and operations you performed before the problem occurred, as well as any steps taken to resolve the problem. The more complete and concise the description is, the more effective Visa Technical Support will be in assisting you.
- The exact text, number, and source of all error messages related to the problem.

- Information about the Visa Clearing Exchange application, such as the release version, specification file version, table file versions, and server operating system. For easy access to this information, see the About Page, which is found by navigating from **Help > About**.
- Any other information that could be pertinent or useful in resolving the problem. For example, if the problem involves file transfers, your EAS version and IP address could be essential to resolving the problem.

VCX Documentation

This documentation is about, or in relation to, Visa Clearing Exchange (VCX).

VCX Documentation

- *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Client Implementation Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Messages Reference Guide*
- *BASE II Clearing Visa Clearing Exchange (VCX) Reports Reference Guide*

BASE II Documentation

- *BASE II Clearing Data Codes*
- *BASE II Clearing Interchange Formats, TC 01 to TC 49*
- *BASE II Clearing Interchange Formats, TC 50 to TC 92*
- *BASE II Clearing Overview*
- *BASE II Clearing Services*

VCMS Testing Documentation

- *VisaNet Certification Management Service (VCMS) BASE II Testing Guide*

Other Documentation

- *Visa Core Rules and Visa Product and Service Rules*
- *Visa Europe Operating Regulations – Processing*
- *Visa Merchant Data Standards Manual*

Chapter 2

Primary Features in VCX

Visa Clearing Exchange (VCX) software has been created to assist clients with sending, receiving, and processing clearing files.

VCX includes these features:

- Windows OS
- Linux OS
- Client-Server architecture
- Security framework
- Browser-based user interface
- Flexible reporting options
- Batch procedure customization via run control options

VCX is currently supported on Windows and Linux operating systems only and is based on a client-server architecture. Users are able to use standard Google Chrome and Microsoft Edge browsers to access the software. The application offers an intuitive user interface, consistent user and system administration flows, and streamlined run control options.

VCX Key Features

Visa Clearing Exchange (VCX) provides flexibility in configuring the application and daily processing.

Key Features:

- System administration
- Client-server architecture
- Format-independent architecture
- Use of rules engine and value tables
- Run control options
- Secure file transfer support
- Run automation

Format-Independent Architecture

VCX software uses a common internal format for its basic core processing, validation, and reporting. VCX fully supports processing of BASE II transactions.

Rules Engine and Value Tables

VCX transaction validation is performed using a common internal format and business rules defined in rules and value tables. The business rules and value tables have an effective date that supports the testing of future business enhancements by setting the table date to the effective date of the release. When new value tables and rules are developed, the changes will be distributed via TC 54 – Visa Clearing Exchange Table Update Record Transactions and will be automatically applied to the endpoint's environment. There will be no endpoint action required. If changes to rules and value tables are not received, they can be requested from Visa to the processing center through incoming interchange via TC 54.

VCX Tables include: Account Range Definition (ARDEF) table, Visa Identifier (VID) table, value tables, and rules tables.

Note: Endpoints that choose not to use VCX but want to receive VID, ARDEF, and other value table updates can choose to implement the TC 54 – Visa Clearing Exchange Table Update Record Transactions.

ARDEF Table:

The ARDEF Table contains all valid ARDEF entries, namely: the account prefix range, its associated issuing identifier, card-number length indicator, check-digit indicator, product ID, and account funding source. The ARDEF file is used to verify account numbers received in interchange.

VID Table:

An acquirer ID and issuer ID are six-digit system number numerics used by Visa to identify endpoints. The VID table contains all valid acquirer and issuer IDs and their BASE II processing status codes. The VID table file is used to associate acquirer and issuer identifiers in interchange with their account numbers received in interchange.

Value Tables:

The repository value tables are made up of various tables used in the validation of BASE II transactions. Examples include:

- Country Code
- Currency Code
- Merchant Category Code (MCC)
- Fee Reason Code
- Return Reason Code

Rule Tables:

The Rules tables contain business validation logic used by VCX to validate transaction data. The rules table is internal to VCX processing and layouts for this file are not provided.

Automatic Table Updates

Visa can send updates required to keep the endpoint's repository tables current on a daily basis. If an endpoint's repository tables are out of date, repository table update transactions sent by Visa can bring the tables up to date. The VCX software will determine and create the table updates required to bring processing center tables up to date, based on the current version of the tables at Visa and the endpoint's current table versions.

Note: The processing center must run the VCX outgoing process to provide Visa with the current status of its tables in order to receive table updates tailored to its current table levels. Otherwise, the default set of table updates will be sent.

Run Control Options

The use of run control options allows processing centers to customize VCX software to their specific processing needs. Users can create various run control profiles to satisfy their daily processing needs, such as testing, production, and special runs.

Run control options allow the selection of various functions, including, but not limited to:

- Establishing duplicate batch detection settings.
- Splitting incoming ITF files into multiple CTF files.
- Setting tolerance checking limits at the item levels.

See the *Run Options* chapter for a complete list of supported run control options.

Configurable Parameters within the Config File

Users can use the Config file to configure various parameters, like customizing archive folders, setting user and password expiration times, and archiving and retention periods for files and logs. For details on the parameters, see the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

System Administration

The administrative function provides configuration tools for these functions:

- User Setup – Establishes administrator and operator IDs, passwords, operator privileges, approving operator IDs, resets passwords.
- BASE II CIB Setup – Set up CIBs for test and production.
- File Transfer Setup – Set up file transfer interface connection to VCX via Direct Exchange (DEX) Open File Delivery (OFD), Extended Access Server (EAS), or Visa File Gateway (VFG).
- Batch Procedure – Set up and maintain batch procedures.

VCX Administrator Responsibilities

Processing centers with multiple VCX users must designate one or more persons as the system administrator for VCX, Release 1.0. The system administrator will have these responsibilities:

- Ensure that there are at least two administrators for VCX, Release 1.0
- Managing a multi-operator environment
 - Adding and removing operator IDs
 - Designating operator modes and access level
 - Viewing all operators and administrators for their installation
- Storing a VCX application key

Note: The VCX application key must be securely stored in case the VCX administrator cannot log in to VCX. If the application key is not available to restore access, the VCX software must be completely reinstalled.
- Setting up a secure interface for EAS, DEX OFD, or VFG
- Setting up BASE II CIB data folders
- Ensuring that run control options are set correctly
- Ensuring that table and file updates are performed in a timely manner by monitoring the dashboard that contains the table and file update information on a regular basis
- Mentoring new operators
- Assisting operators to resolve problems

VCX Operator Responsibilities

A VCX operator may perform these tasks:

- Initiating incoming edit run and interchange transaction files (ITFs) from VisaNet
- Setting up temporary run control options that override permanent options for a single processing run
- Facilitating reprocessing of any transactions that were returned from VisaNet and needed to be validated by a subsequent outgoing edit run
- Receiving the outgoing center transaction files (CTFs) from the processing center's pre-edit software so VCX can validate the information during an outgoing edit run
- Facilitating reprocessing of any transactions rejected by the outgoing edit run and resubmitting them into outgoing edit run
- Preparing outgoing interchange for transmission to VisaNet
- Reviewing or printing reports

Formatting

One of the functions of VCX is to format CTF data into ITF data during an outgoing cycle and to format ITF data into CTF data during an incoming cycle. The processing center controls some formatting functions so that incoming files and transactions can be easily processed by the center's internal systems.

Validation

Validation of interchange files is a major function of VCX. In addition to standard validation, processing centers can select additional forms of validation, such as the setting of tolerance levels at the item levels and duplicate file and batch detection. Additional validation functions are controlled through run control options.

Reporting

VCX provides various reports. Formats for dates, time, and numeric values can be modified according to national standards and are available using run control options.

For details on the reports, see the *BASE II Clearing Visa Clearing Exchange (VCX) Reports Reference Guide* or the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

Reports generated by VCX will be identified by the prefix VX, followed by a three-digit number.

Report Types

Report Series	Report Type
100–199	Outgoing reports
200–299	Incoming reports

Report Types

Report Series	Report Type
300–399	On-request reports
700–799	Optional transaction-based reports
999	Index of reports

For reports that provide totals at different levels, the report ID will include an alphabetic extension to represent the totaling levels:

- A – Transaction detail level
- B – Batch/acquirer/issuer ID level
- C – File/center level
- D – Central Processing Date (CPD) level
- E – Run level
- F – Cycle level

Secure File Transfer Support

BASE II interchange files can be securely sent and received to and from Visa, using these delivery channels:

- DEX OFD – Visa will provide the option for endpoints to use File Transfer Protocol Secure (FTPS) to transfer files to DEX OFD for VCX, Release 1.0. The FTPS Client for OFD is included with the VCX software installation package. Acquirers and Issuers may use this software for secure file transfer on outgoing runs. It is still required for endpoints to use third-party secure FTP server software for incoming runs through OFD. Additionally, FTPS products that are Requests for Comments (RFC) 4217-compliant and support a command line (scripting) interface and Transport Layer Security (TLS) 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from OFD.

See *Visa Direct Exchange Open File Delivery Service Implementation Guide* for information about channel encryption.

- EAS – Visa will provide the option for endpoints to use an FTPS connection to an EAS for VCX, Release 1.0. The FTPS Client for Extended Access is included with the VCX software installation package. Acquirers and issuers may use this software for secure file transfer. Additionally, FTPS products that are RFC 4217-compliant and support a command line (scripting) interface and TLS 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from EAS.

See *Visa Extended Access Server Endpoint Guide* and *Visa Extended Access Server Centralized (EAS-C) Client Implementation Guide* for information about EAS setup.

- VFG – Visa will provide the option for endpoints to use an FTPS to VFG for VCX, Release 1.0. The FTPS Client for VFG is included in the VCX software installation package. Acquirers and issuers may use this software for secure file transfer on outgoing runs. It is still required for endpoints to use third-party secure FTP server software for incoming runs through VFG. Additionally, FTPS products that are RFC 4217-compliant and support a command line (scripting) interface and TLS 1.2 or later can be configured to work with VCX, Release 1.0.

VCX will support the standard file compression used to send and receive compressed ITF files to or from VFG.

See *Visa File Gateway Client Implementation Guide* for information about VFG setup.

- Visa Cloud Connect – Support for Cloud Connect will be announced later.

Note: DEX OFD delivery channel will be replaced by a new Visa File Gateway (VFG) in future releases.

BASE II TC 54 File Delivery

Acquirers and issuers may receive the TC 54–Visa Clearing Exchange Table Update Record Transactions in either UNDIFF file type or by using BASE II customized file delivery. Direct Exchange, Visa File Gateway (VFG), or an Extended Access (EA) server can be used for either file delivery method.

The **VCX54** customized delivery file type enables the TC 54–Visa Clearing Exchange Table Update Record Transactions delivery as a custom file type. This allows both Visa Clearing Exchange (VCX) users and non-VCX users to receive the TC 54–Visa Clearing Exchange Table Update Record Transactions in a separate file.

BASE II Customized Delivery Type Field Value

BASE II customized delivery files are identified by a unique value in the BASE II Customized Delivery Type field, positions 109–113, of TCR 0–Incoming ITF records, in TC 90–Header Records.

Direct Exchange, VFG, and EA Server File Type Value

File type values identify the kinds of files that are listed on the available incoming file download lists for Direct Exchange, VFG, and EA Server file transfers.

The file type of **VCX54** identifies files containing TC 54–Visa Clearing Exchange Table Update Record Transactions.

Tools

- Initialize history files – used to initialize the history files. See *History File*.
- EAS File Services (EFS) tool – used to view ready or purge files under a specific EAS profile and BASE II CIB. Additionally, users can cancel ready files or restore purge files. If the user's cancel ready or restore purge files request fails, the errors can be viewed in the file transfer protocol (FTP) log.
- View Application Log tool – lists all actions completed by the user in VCX as well as their timestamps.

Batch Procedures

Users can set up and maintain batch procedures, submit them for immediate execution or schedule for a given date or time using the scheduler, and view completed jobs and batch job history. Users can create multiple batch procedures. The View Batch Logs function allows users to view the log after the batch procedure is run. Users will be able to view, edit, duplicate, or delete the current batch procedures that can be run or create a new batch procedure. Batch procedures define the set of parameters for a job that users will run. A batch job is the execution of the batch procedure.

Users will be able to automate the execution of the batch procedure by running a script on a Linux or Windows server. The script can call a scheduler to run a batch at a specific date and time.

Batch procedures may be used for these processes:

- Incoming File Transfer (EAS, DEX OFD, or VFG)
- Incoming Run
- Outgoing Run
- Outgoing File Transfer (EAS, DEX OFD, or VFG)
- Table Extracts
- Table Reports

A single batch procedure can include a combination of these processes.

Note: VCX will not be backwards compatible with any previous version of VCX or Edit Package.

History File

The History File is used to store the history of the outgoing and incoming processing runs, and to control reruns and assign batch numbers for multiple daily runs.

Software Distribution and Updates

The two primary distribution events for Visa Clearing Exchange (VCX) are for the initial installation, and subsequent updates.

Initial Software Distribution

The installation packages for Visa Clearing Exchange (VCX) will be available as downloads from Visa Access. Visa Access is the current distribution method for most software updates.

The installation packages on Visa Access will contain the latest version of the software package and Repository files.

To download installation packages from Visa Access, processing centers are expected to have file transfer capabilities from Visa Access to their Host system. The installation packages will contain instructions for installation.

Business Rules Updates

Updates for business rules, Visa Identifiers (VIDs), and account range definitions (ARDEF) will be received and automatically applied during an incoming run.

After the initial installation of Visa Clearing Exchange (VCX), updates to business rules will be contained in the TC 54 transactions that will be included in the incoming interchange files.

Updates to business rules and Repository files will include an effective date on which the updates will automatically become active.

Note: VCX will generate a return code value of 5 when a file processed by an incoming run does not contain any TC 54 transactions. VCX will generate a return code value of 6 when the update process during an incoming run is not successful. A return code value of 6 may also be generated along with an appropriate message in the processing log to indicate that the software components required for a successful update are available through Visa Access.

Software Updates

Software updates will be available on Visa Access for Visa Clearing Exchange (VCX).

Chapter 3

System Requirements for VCX

This chapter includes the minimum system and operational requirements for the installation of Release 1.0 of the Linux or Windows version of Visa Clearing Exchange (VCX) in the test or other nonproduction environment. For the production environment, disc space requirements in particular will be dependent on how VCX, Release 1.0 is customized.

VCX, Release 1.0 will be supported on Linux and Windows operating systems. Visa cannot guarantee the functionality of VCX, Release 1.0 on unsupported software, hardware, and operating systems. To ensure full support from Visa, endpoints should use VCX, Release 1.0 on one of the approved software, hardware, and operating systems. Endpoints must migrate to a supported operating system.

System Requirements

Before installation, Visa Clearing Exchange (VCX) administrators should check that every server on which VCX, Release 1.0 is to be installed has the necessary system and operational requirements. See the *VisaNet Global Technical Letter and Implementation Guide* or your Visa Representative for the most recent supported versions system requirements for VCX.

Requirements include:

- Hardware requirements
- Memory requirements

- Disk space requirements
- Operating system requirements
- Run time library requirements
- Client browser requirements

System Requirements for VCX

Before installing Visa Clearing Exchange (VCX), clients must ensure that they can meet the necessary system and operational requirements.

Important: The most up-to-date system requirements for VCX are available in the most recent version of the *VisaNet Business Enhancements Global Technical Letter and Implementation Guide*.

Installation Hardware Requirements:

These are the minimum hardware requirements for installing VCX:

- **CPU** – 4CPU/4 CORE, 64bit
- **Disk storage** – 500 GB
- **Memory** – 32 GB
- **Communication Protocol** – HTTPS

Note: VCX only supports HTTPS communication between client and server machines. This requires a TLS certificate. During part of the installation process, you will have an option to install self-signed certificates generated by the OpenSSL application or a certificate issued by any recognized certificate issuing provider.

- **File Transfer Protocol** – FTPS

Operating System Requirements:

Visa supports the VCX application running on Linux OS or Windows OS:

- **Red Hat Enterprise Linux** – RHEL 8.0 (minimum) or 9.4 (maximum)¹
- **Windows Server 2022** – Standard Core

¹ VCX is supported on the listed version and may be supported on a higher version provided the higher version maintains backward compatibility with the listed version.

Runtime Library:

VCX requires these libraries, which the Automated Prerequisite Software Installer installs:

- **NodeJS** – Version 20¹
- **Open JDK/Java** – Version 21¹
- **Open SSL** – 1.1.1¹

Browser Requirements:

VCX can be used with these browsers:

- **Google Chrome** – Version 120¹ (minimum) or Version 130 (maximum)
- **Microsoft Edge** – Version 120¹ (minimum) or Version 130 (maximum)

Databases:

VCX requires this database with a generic setup:

PostgreSQL – Version 16.4¹

Note: PostgreSQL setup will be included in the Prerequisite Install script.

S-Nail:

By default RHEL 9.0 or above uses S-Nail

Note: S-Nail replaces Mailx from RHEL 9.0 or above

VCX requires the use of Mailx for two-factor authentication (2FA) for Linux:

Mailx–Version 12.5 or above (this is the default version with Linux 8 or above)

SMTP Mail:

VCX requires the use of SMTP mail for multifactor authentication (MFA) for Windows:

Linux File Transfer Program (LFTP):

VCX requires the use of LFTP if using Linux:

LFTP – Version 4.4.8 or above (by default on Linux)

File Transfer Program (FTP):

VCX requires the use of FTP for Windows:

¹ VCX is supported on the listed version and may be supported on a higher version provided the higher version maintains backward compatibility with the listed version.

FTPS – or File Transfer Protocol Secure is a secure file transfer protocol that uses encryption to secure file transfers over a network using Transport Layer Security (TLS) protocol.

PORT:

Ensure that ports 3000 and 8000 are open and available for VCX services to run.

More Information

Contact a regional Visa representative or your client support team to learn more.

Chapter 4

Run Control Options in VCX

Visa Clearing Exchange (VCX), Release 1.0 includes numerous run control options that allow processing centers to customize VCX to their needs. For VCX, Release 1.0, the run control options have been significantly revised. A default set of run control options for incoming and outgoing settings will be supplied with the VCX, Release 1.0 software. Endpoints must choose permanent run control options, data entry options, and report archiving options after the first time the main application is run after installation. Permanent settings, options, and table overrides can be changed at any time.

Note: For current Edit Package, Release 4.0 users, endpoints should review their current Run Control Options from Edit Package, Release 4.0, so they can replicate the run control option customization that they have in VCX. For complete information regarding run control options, see the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

Incoming Processes Controlled by Run Control Options

This describes, at a high level, the types of incoming run control options that are available in Visa Clearing Exchange (VCX), Release 1.0.

Types of Incoming Run Control Options

Incoming Function	Purpose
Identify Processing Center	Defines the processing center's BASE II CIB, Name, and Security Code.
Perform Rerun Processing	Indicates that a rerun is to occur.
Verify Run Date for re-processing incoming files	Verifies the central processing date in the ITF header when files are being reprocessed.
Set Effective Date for VCX Tables	Specifies effective date used by VCX to establish which table entries to use during processing.
Validation Exception Report Options	Specifies sequencing and formatting of reports.
Identify Run Mode	Specifies whether the interchange to be processed contains production or test data.
Set History Maintenance Parameters	Specifies the number of calendar days that run, file, and batch data are retained on the VCX History File.
Select Report Date, Time, and Numeric Values Format	Specifies format for date, time, and numeric values for reports.
Create Incoming CTF Header Record	Creates a TC 90 header record on all incoming CTFs.
Split Incoming ITF into Multiple CTFs	Splits an incoming ITF into separate CTFs based on account ranges, destination IDs, and transaction codes.
Create Returned Items File	Writes transactions returned by the Visa Interchange Center (VIC) on incoming interchange to a separate file in a specified format.
Pass transactions to CTF	Controls which transactions are written to the CTF
Generate Transaction-Based Reports	Indicates whether incoming transaction-based reports are to be generated.
Override Table File Entries	Used to modify table files during a run.

Outgoing Processes Controlled by Run Control Options

This describes, at a high level, the types of outgoing run control options that are available in Visa Clearing Exchange (VCX), Release 1.0.

Types of Outgoing Run Control Options

Outgoing Function	Purpose
Identify Processing Center	Defines the processing center's BASE II CIB, security code, and name.
Reprocess Outgoing CTF Files	Identifies the run to be reprocessed.
Set Run Date	Identifies the VCX processing date, overriding the system date.
Set Effective Date for VCX Repository Tables	Specifies the effective date used by VCX to establish which table entries and rules to use during processing.
Validation Exception Report Options	Specifies the sequencing and formatting of the reports.
Indicate Run Mode	Specifies whether the interchange to be processed contains production or test data.
Set Size Limits for Batch	Specifies the maximum number of TCR records to be written to a single batch.
Set History Maintenance Parameters	Specifies the number of calendar days that run file and batch data are retained on the VCX History File.
Detect Submission of Duplicate Batches	Identifies a duplicate batch and issues a warning or error message to alert the user.
Set File Identification Modifier Character Used in the Unique File Identifier	Can be used to specify a single character included in the BASE II Unique File ID filed on the File Header Record. Files can be readily recognized by the Unique File ID modifier.
Define File Type	Specifies File Type populated in the header record file type field.
Set Item Tolerance Levels	Specifies the amount of tolerance levels for individual transactions by currency.
Set Reject Limits	Sets the ITF reject limit as a percentage of transactions that are in error.
Create Rejected Items File	Writes transactions rejected by VCX on an outgoing run to a separate file in the specified format.

Types of Outgoing Run Control Options

Outgoing Function	Purpose
Override Table File Entries	Used to modify table file entries during a run.
Generate Transaction-Based Reports	Indicates whether outgoing transaction-based reports are to be generated.

Chapter 5

VCX Validation Messages

During incoming edit runs, Visa Clearing Exchange (VCX), Release 1.0 performs file integrity edits to ensure that incoming files have not been corrupted or changed during transmission. Checks are also performed for duplicate incoming files.

During outgoing edit runs on VCX, VCX validates transaction data to ensure that the data is complete and correct. Transactions are subject to several stages of validation.

Optional edits can also be invoked by permanent or temporary run control options to:

- Check for duplicate batches
- Check for transaction reject counts that exceed error limit percentages

When an error is detected during an incoming run, a validation message number and a brief text are displayed on VCX reports.

VCX validation messages are very specific and consist of these:

- Message Number: V0011
- Message Text: BANK NAME MUST NOT EQUAL SPACES
- Message Description: The bank name must be a left-justified 25-character field and must be space-filled to the right.
- Message Action: Enter the correct value for the field.
- Message Transaction Type: Sales Draft—All

For more detailed explanations and examples of VCX Validation Messages, see the *BASE II Clearing Visa Clearing Exchange (VCX) Messages Reference Guide*.

Chapter 6

VCX Implementation Considerations

To implement Visa Clearing Exchange (VCX), Release 1.0, acquirers, issuers, and processors should consider the implementation schedule, installation, computer resources, and the impacts to their internal system interfaces as a part of the migration and implementation strategy. The same BASE II CIB cannot be used for VCX and Edit Package 4.0. You can have a CIB unique to either VCX or Edit Package 4.0, but not both.

Note: To install VCX, acquirers, issuers, and processors must follow the detailed instructions provided in *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*.

For implementation, acquirers, issuers, and processors should consider the following, before installing VCX, Release 1.0 software:

1. Download software installation package
2. Installation System Requirements
3. Implementation Strategies
4. Key Implementation Tasks
5. Key Project Dates
6. Service Roles and Responsibilities
7. Software and Table Updates

Download Software Installation Package

The Visa Clearing Exchange (VCX) software installation package can be downloaded from Visa Access. Endpoints must ensure they have file transfer capabilities from Visa Access to their Host system.

Note: Clients in some countries may require prior approval to use VCX. Contact your regional Visa support representative for further detail.

System Requirements

See *System Requirements for VCX* chapter earlier in this manual for the minimum system and operational requirements for the installation of Visa Clearing Exchange (VCX), Release 1.0 in the test or other nonproduction environment. For the production environment, disk space requirements in particular will be dependent on how VCX is customized.

Implementation Planning

Endpoints must begin the installation planning process with sufficient lead-time in case you must upgrade hardware or operating systems, and expand disk, or memory size allocations, as a result of the system requirements for Visa Clearing Exchange (VCX), Release 1.0.

Key Implementation Tasks

Processing centers can obtain the Visa Clearing Exchange (VCX), Release 1.0 software by downloading the VCX installation package from Visa Access.

Installation of the software, internal testing, and VCMS testing may begin as soon as the project is initiated. Visa recommends that endpoints successfully complete internal testing and VCMS testing before implementing the Linux or Windows version of VCX, Release 1.0 into the production environment.

As every processing center is different, processors should consider these processing requirements before implementing VCX, Release 1.0:

- Run control options, scheduling, and procedures.
- Internal system interfaces with VCX for new reports and input and output file names.

- Report distribution systems or processes to accommodate the report changes.
- Internal procedures and staff training for VCX.

Key Project Dates

Discuss with your regional Visa representative for the key dates and schedule for Visa Clearing Exchange (VCX).

- VCX available for download on Visa Access
- VCMS Testing period
- Production activation of VCX

Service Roles and Responsibilities

This describes the roles and responsibilities for implementing Visa Clearing Exchange (VCX), Release 1.0.

Processing Centers

The role of the processing center is to set up and perform the installation and implementation of VCX, Release 1.0 in this order:

1. Download the installation package from Visa Access.
2. Install VCX software and verify successful installation.
3. Review and customize run control profiles and set up options according to your processing requirements.
4. Contact your Visa representative to schedule and coordinate VCMS testing and to request VCX VCMS Full File Replacement.
5. Perform VCMS testing using VCX.
6. Perform internal testing using production-like data.
7. Perform interface testing between VCX and your internal systems.
8. Contact your Visa representative to schedule an implementation date.
9. Implement VCX in production environment.

Visa Regional Client Support Representatives

The role of a Visa regional client support representative is to assist processors to set up and perform the installation and implementation of VCX, Release 1.0 as follows:

1. Provide a link to Visa Access with the VCX installation package.
Note: Some countries may require prior approval to implement VCX software. Contact Product Office for a list of countries.
2. Request VCMS VCX Full File Replacement (FFR), or endpoints can request this file.
3. Contact VCX Support for implementation-related issues encountered by processing centers.
4. Inform VCX Support of a processing center's implementation, going-live date.

Software and Table Updates

Endpoints should always confirm that the most recent software and whether for the initial installation, testing, or production processing of Visa Clearing Exchange (VCX), Release 1.0. The latest software version can be found on the VCX page in Visa Access.

The initial installation package and subsequent software updates for VCX will be available only through Visa Access. Test VCX tables, including repository, VID, and ARDEF tables, can be requested during the testing period via TC 54 through VCMS. Production VCX Full File Replacement (FFR) must be requested before production activation.

Chapter 7

Implementation Steps - VCX for Linux and Windows

This chapter contains information for implementing Visa Clearing Exchange (VCX) on a Linux or Windows system.

These steps must be completed to ensure proper implementation of VCX.

- Conduct required VCMS testing of VCX before implementation.
- Complete these implementation tasks successfully:
 - Prepare the last outgoing CTF for Edit Package, Release 4.0 (for current Edit Package, Release 4.0 users)
 - Prepare the production environment for VCX
 - Process the first incoming ITF using VCX
 - Process the first outgoing CTF using VCX

Note: Implementation of VCX must take place on the date agreed upon with a Visa Regional client support representative. Allow at least ten business days between confirmation of successful VCMS testing and the implementation of VCX to allow sufficient lead time for updating BASE II tables.

Install VCX

The first step in implementing Visa Clearing Exchange (VCX) is to install the system.

Users of VCX, whether on Linux or Windows, should follow the software installation instructions in *BASE II Clearing Visa Clearing Exchange (VCX) User Guide*. The installation instructions are defined in detail for each operating system.

Pre-VCMS Testing Activities

Preparing the Visa Clearing Exchange (VCX) environment for BASE II VCMS testing includes steps provided below. To ensure successful VCMS testing, it is important to carefully follow these steps. This task contains the steps that must be completed before VCMS testing for an outgoing edit run.

Pre-VCMS Testing Tasks for VCX

1. Install VCX in the test environment.
2. Configure your application:
 - a. Set the run control options and profile that is to be used in the production environment.

Note: A migration tool is available for PC Edit Package 4.0. The migration tool can be used to generate the Edit Package, Release 4.0 settings, configurations, and a summary report that can be used as a reference while configuring VCX.

- b. Set up the BASE II CIBs.
 - c. If applicable, set the Extended Access Server (EAS) or Direct Exchange (DEX) Online File Delivery (OFD)/Visa File Gateway (VFG) Interface.
 - d. Define an incoming batch procedure with the run mode set to **TEST**.

Note: See *BASE II Clearing Visa Clearing Exchange (VCX) User Guide* for details on configuration.

3. Prepare Files:
 - a. Initialize the history files, if applicable, from the Tools tab.

Load VCMS VID and ARDEF Tables

The current VCMS tables must be loaded before VCMS testing. These include VID and ARDEF information from the latest BASE II updates.

Contact your Visa regional client support representative to arrange for the current VCMS tables to be delivered either by transmission to the Extended Access Server (EAS) or by Direct Exchange (DEX) Online File Delivery (OFD)/Visa File Gateway (VFG). This should be done

immediately before your scheduled VCMS testing date so that the VID and ARDEF information is current.

The VCMS testing files include complete the VCX Full File Replacement of tables and rules files previously loaded from the installation package.

1. Load Current Tables:
 - a. Arrange for the latest table files to be sent to an EAS or DEX OFD/VFG.
 - b. Execute an incoming run using the new table files as input.
 - c. Check and verify the results.

The VCMS testing replacement files are loaded by executing an incoming batch procedure.

The tables will be successfully uploaded when the return code is a value of **0** (zero).

Prepare Files for VCMS Testing

The Visa Clearing Exchange (VCX) history file should be initialized before VCMS testing to eliminate any entries generated during installation and testing.

To initialize history files, users should select **Tools > Initialize History Files**, then select the BASE II CIB you want to initialize the history files for.

Create Run Control Option Profiles

There are many run control options that can be used to customize Visa Clearing Exchange (VCX) for your processing center. For VCMS testing, simply use the run control options you intend to use for production.

See the *BASE II Clearing Visa Clearing Exchange (VCX) User Guide* for details on how to create run control profiles.

Verify Processing Center Information

The outgoing VCMS ITF file that will be produced from an outgoing VCX run will be collected into the test BASE II system. The file must be identified with your processing center's BASE II CIB and security code.

Ensure that the correct values for the BASE II CIB and security code fields have been entered. The name of your institution should be entered in the Name field of the BASE II CIB Setup page, so that it will appear on report headings.

VCMS Testing for VCX

Testing should verify that Visa Clearing Exchange (VCX) functions properly within the internal Host system. Endpoints should consider how they are currently using Edit Package and all of its reporting capabilities.

The VCX testing should address the processing of outgoing and incoming files. Testing should be designed to ensure that:

- The endpoint can receive and process incoming interchange files with VCX.
- The endpoint's outgoing files created with VCX will be accepted and processed by the BASE II component of the VisaNet Certification Management Service (VCMS).
- The endpoint can process and send outgoing interchange files without receiving unexpected returns or reclassifications.
- The VCX reports meet the requirements of the endpoint's operations area.
- The VCX history and batch reconciliation work properly.

For all VCX endpoints, contact your Visa representative for setting up the test environment. For testing, Visa requires these activities:

- The endpoint must be connected to VCMS and its BASE II CIB must be activated for testing.
- The VCX Questionnaire must be completed for each endpoint to be activated in VCMS before the start of testing.

All endpoints, regardless of platform, must follow the required testing procedures.

VCX testing should be conducted in these steps:

- Test using the test data provided with the installation package.
- Test using production-like data
- Test with Visa using VCMS.

The table describes the overall responsibilities for endpoints and Visa staff when conducting VCMS BASE II Testing for VCX.

Testing Tasks for VCX

Step	Task	Description	Party Responsible	Completed?
1	Install VCX in the test environment	Install VCX in the test environment.	Endpoint	
2	Configure your test VCX environment	Define your: - Processing Center's Identification settings (BASE II CIB) - File Transfer Interface - Run Control profiles to be used in the test environment	Endpoint	
3	Configure Batch Procedures for the incoming run and file transfer	Configure batch procedures for the incoming file transfer and incoming run in Test mode using BASE II CIB, file transfer interface, and run control profile configured in the previous step.	Endpoint	
4	Load current tables	- Arrange for the latest table (VCMS Full File Replacement (FFR)) files to be sent online/offline from VCMS to an Extended Access Server (EAS) or Direct Exchange (DEX) Open File Delivery (OFD)/Visa File Gateway (VFG). - Execute the Batch procedure configured in previous step using the FFR table files as input. - Check and verify the results.	Endpoint	
5	Request test data	Request test data using the visa Test System-Clearing and Settlement (VTS-CS)	Endpoint	
6	Process test data	Process VCMS BASE II test data that accurately reflects a mix of production activity.	Endpoint	
7	Validate test files.	Ensure that the test files are staged, with TEST in the header, on the Extended Access Server (EAS) or Direct Exchange (DEX) Online File Delivery (OFD)/Visa File Gateway (VFG) for VCMS collection.	Endpoint and Visa	
8	Configure outgoing batch procedure	Configure additional batch procedures for outgoing run and file transfer in TEST mode.	Endpoint	
9	Prepare VCMS outgoing test data	Prepare VCMS outgoing test data. Note: Endpoints must only use test account numbers for VCMS testing.	Endpoint	
10	VCMS testing	Perform VCMS testing by executing the outgoing batch procedure configured in step 8.	Endpoint	

Testing Tasks for VCX

Step	Task	Description	Party Responsible	Completed?
11	Verify test results	Verify (and explain) any returned or rejected items. Ensure counts and amounts are correct.	Endpoint and Visa	
12	Contact Visa, if necessary	Validate and test results from VCMS BASE II testing.	Endpoint	

Test Installation Package

To verify that the system was installed correctly, the endpoint must execute an outgoing Visa Clearing Exchange (VCX) run and an incoming VCX run using the supplied test files.

Endpoints can validate results of the test VCX runs by reviewing generated VCX reports.

Test Data

For information on VCMS testing, see the *VisaNet Certification Management Service (VCMS) BASE II Testing Guide* to request or send data to or from the VCMS environment.

Ongoing Testing

BASE II processing rules change with every business release, and often with off-release code installations. As a result, Visa strongly recommends that endpoints retest their internal processing rules and their incoming files for Visa Clearing Exchange (VCX) through VCMS for each release, as well as for off-release installations.

See the *VisaNet Certification Management Service (VCMS) BASE II Testing Guide* for information on testing approaches for endpoints to test outgoing CTFs and incoming ITFs.

New Client Production Implementation Steps for VCX

This describes the preparation steps for implementing Visa Clearing Exchange (VCX) in the production environment for new clients. Prior to following these checklists, complete the VCX

Questionnaire with information on your BASE II CIBs, production effective date, and operating system.

Note: Contact your Visa Representative for information on subscribing to Custom File Type for TC 54 VCX Record Update Transactions. This list is illustrative only and may not be comprehensive. It does not replace a client's own assessments of necessary steps.

Implementation Checklist (New VCX Client)

Step	Task	Description	Party Responsible	Completed?
1	Prepare the VCX environment	Install VCX in the production environment.	Endpoint	
2	Configure your production VCX environment	Define your: - Processing center's identification settings (BASE II CIB) - File transfer interface - Run control profile that is to be used in the production environment	Endpoint	
3	Configure batch procedures for the incoming run and file transfer	Configure a batch procedure for the incoming file transfer and incoming run in production mode using the BASE II CIB, file transfer interface, and run control profile configured in the previous step.	Endpoint	
4	Request VCX FFR	Request the VCX Full File Replacement (FFR) prior to running your first incoming batch procedure.	Endpoint	
5	Execute the initial VCX incoming batch procedure.	- Execute the batch procedure configured in step 3. Use your first VCX incoming file that contains the VCX FFR as input. - Check and verify the results. Verify that the VX-299 report indicates successful completion of the incoming run and the return code is a value of 0.	Endpoint and Visa	
6	Configure outgoing batch procedure.	Configure a batch procedure for outgoing run and file transfer in production mode.	Endpoint	
7	Execute initial VCX outgoing batch procedure.	- Execute the first outgoing run for VCX. Check and verify the results of the run. - Verify that the VX-199 report indicates successful completion of the outgoing run and that the return code is a value of 0. - Validate successful transfer of the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II.	Endpoint and Visa	

Existing Client Production Implementation Steps for VCX

This describes the preparation steps for implementing Visa Clearing Exchange (VCX) in the production environment. Prior to following these checklists, choose your production cutover option and complete the VCX Questionnaire with information on your production cutover option, BASE II CIBs, production effective date, and operating system.

1. Cutover CPD incoming file includes the production VCX Full File Replacement (FFR), TC 44 Batch Acknowledgment, and Clearing Data. This option is fully automated and does not require additional action on the part of the endpoint, but you will not be able to generate your last Edit Package, Release 4.0 (EP 4.0) EP-220 Batch Acknowledgment report correctly.
2. Cutover CPD incoming file will not include VCX FFR. This option requires requesting the VCX FFR separately, but allows you to generate the last Edit Package, Release 4.0 (EP 4.0) EP-220 Batch Acknowledgement report correctly.
3. Subscribe for the VCX custom file type to automatically receive VCX FFR in a separate file. This option will give you the ability to generate an EP-220 report using Edit Package, Release 4.0 (EP 4.0) and process the VCX FFR to prepare for the first day of processing. To subscribe to the TC 54 VCX Table Record Update Transaction custom file type, please contact your Visa representative.

Note: This list is illustrative only and may not be comprehensive. It does not replace a client's own assessments of necessary steps.

Implementation Checklist (Option 1 – TC 54 Records are Included in the UNDIFF File)

Step	Task	Description	Party Responsible	Completed?
1	Execute final Edit Package, Release 4.0 outgoing run.	• Execute an outgoing run using Edit Package, Release 4.0. • Transfer ITF to an Extended Access Server (EAS) or Direct Exchange (DEX) Online File Delivery (OFD)/Visa File Gateway (VFG) • Verify that ITF is collected by BASE II • Preserve Edit Package, Release 4.0 software and files	Endpoint	
2	Prepare the VCX environment	Install VCX in the production environment.	Endpoint	

Implementation Checklist (Option 1 – TC 54 Records are Included in the UNDIFF File)

Step	Task	Description	Party Responsible	Completed?
3	Configure your production VCX environment	Define your: - Processing center's identification settings (BASE II CIB) - File transfer interface - Run control profile that is to be used in the production environment	Endpoint	
4	Configure batch procedures for the incoming run and file transfer	Configure a batch procedure for the incoming file transfer and incoming run in production mode using the BASE II CIB, file transfer interface, and run control profile configured in the previous step.	Endpoint	
5	Execute the initial VCX incoming batch procedure.	- Execute the batch procedure configured in the previous step. Use your first VCX incoming file that contains the VCX FFR as input. - Check and verify the results. Verify that the VX-299 report indicates successful completion of the incoming run and the return code is a value of 0. Note: The VX-220 Batch Acknowledgment report will be generated, but will contain "Not found in history" comment for all outgoing batches processed previously using the Edit Package, Release 4.0 software.	Endpoint and Visa	
6	Configure outgoing batch procedure.	Configure a batch procedure for outgoing run and file transfer in production mode.	Endpoint	
7	Execute initial VCX outgoing batch procedure.	- Execute the first outgoing run for VCX. Check and verify the results of the run. - Verify that the VX-199 report indicates successful completion of the outgoing run and that the return code is a value of 0. - Validate successful transfer of the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II.	Endpoint and Visa	

Implementation Checklist (Option 2 – Request VCX FFR Separately)

Step	Task	Description	Party Responsible	Completed?
1	Prepare the VCX production environment.	Install VCX software in the production environment.	Endpoint	
2	Configure your production VCX environment.	Define your: - Processing center's identification settings (BASE II CIB). - File transfer interface. - Run control profile to be used in the production environment.	Endpoint	
3	Configure a batch procedure of the incoming run and file transfer	Configure a batch procedure for the incoming file transfer and incoming run in production mode using the BASE II CIB, file transfer interface, and run control profile configured in the previous step.	Endpoint	
4	Request current production VCX FFR.	The VCX FFR must be requested at least one day before the production cutover date.	Endpoint	
5	Execute the VCX incoming batch procedure using the VCX FFR as the input.	- Execute the batch procedure defined in step 3, using the new table files received in step 4 as the input. - Check and verify the results. Verify that the VX-299 report indicates successful completion of the incoming run and that the return code is a value of 0.	Endpoint and Visa	
6	Execute the final Edit Package, Release 4.0 outgoing run.	- Execute an outgoing run using Edit Package, Release 4.0. Transfer the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II Central.	Endpoint	
7	Execute final Edit Package, Release 4.0 incoming run.	- Transfer ITF to an EAS or DEX OFD/VFG. - Execute an incoming run using Edit Package, Release 4.0. - Check and verify the results. - Preserve Edit Package, Release 4.0 software and files.	Endpoint	
8	Configure VCX outgoing batch procedure.	Configure a batch procedure for an outgoing run and file transfer in production mode.	Endpoint	

Implementation Checklist (Option 2 – Request VCX FFR Separately)

Step	Task	Description	Party Responsible	Completed?
9	Execute initial VCX outgoing batch procedure.	- Execute the first outgoing run for VCX. Check and verify the results of the run. Verify that the VX-199 report indicates successful completion of the outgoing run and that the return code is a value of 0. - Validate successful transfer of the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II.	Endpoint and Visa	
10	Execute initial VCX incoming batch procedure.	- Execute the batch procedure configured in step 3. Use your first VCX incoming file that contains the VCX FFR as input. - Check and verify the results. Verify that the VX-299 report indicates successful completion of the incoming run and the return code is a value of 0.	Endpoint and Visa	

Implementation Checklist (Option 3 – Subscribe for a Custom File Type)

Step	Task	Description	Party Responsible	Completed?
1	TC 54s	Subscribe to receive TC 54s as a custom file type.	Endpoint	
2	Prepare the VCX production environment.	Install VCX in the production environment.	Endpoint	
3	Configure your production VCX environment.	Define your: - Processing center's identification settings (BASE II CIB). - File transfer interface - Run control profile to be used in the production environment.	Endpoint	
4	Configure the batch procedure for the incoming run and file transfer.	Configure a batch procedure for the incoming file transfer and incoming run in production mode using the BASE II CIB, file transfer interface, and run control profile configured in the previous step.	Endpoint	
5	Load the latest VCX tables.	Use the VCX custom file that contains the VCX FFR to load the latest production tables.	Endpoint	

Implementation Checklist (Option 3 – Subscribe for a Custom File Type)

Step	Task	Description	Party Responsible	Completed?
6	Execute the VCX incoming batch procedure using the VCX FFR as an input.	- Execute the batch procedure defined in step 4 using the new table files as input. - Check and verify the results. Verify that the VX-299 report indicates successful completion of the incoming run and that the return code is a value of 0.	Endpoint and Visa	
7	Execute the final Edit Package, Release 4.0 outgoing run.	- Execute an outgoing run using Edit Package, Release 4.0. Transfer the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II Central.	Endpoint	
8	Execute the final Edit Package, Release 4.0 incoming run.	- Transfer the ITF to an EAS or DEX OFD/VFG. - Execute an incoming run using Edit Package, Release 4.0. - Check and verify the results. - Preserve Edit Package, Release 4.0 software and files.	Endpoint	
9	Configure a VCX outgoing batch procedure.	Configure a batch procedure for an outgoing run and file transfer in VCX in production mode.	Endpoint	
10	Execute initial VCX outgoing batch procedure.	- Execute the first outgoing run for VCX. Check and verify the results of the run. Verify that the VX-199 report indicates successful completion of the outgoing run and that the return code is a value of 0. - Validate a successful transfer of the ITF to an EAS or DEX OFD/VFG. - Verify that the ITF is collected by BASE II.	Endpoint and Visa.	

A representative of the processing center's staff should be on-site or on-call until the outgoing ITF has been collected. If assistance is needed, contact Visa.

Chapter 8

Going Live with VCX

This chapter describes what going live with Visa Clearing Exchange (VCX) entails as well as when to contact support.

Production Implementation of VCX

Implementation of Visa Clearing Exchange (VCX) into the production environment must take place on the date agreed upon with a Visa regional client support representative. Endpoints should allow at least ten business days between confirmation of successful VCMS testing and the implementation of VCX to allow sufficient time to update the BASE II tables.

VisaNet Technical Support

If the client's processing center encounters problems with implementing Visa Clearing Exchange (VCX) in the production environment, the client should report the problem to its Visa regional client support representative, and depending on the severity of the problem, must be prepared to immediately fall back to Edit Package, Release 4.0.

Appendix A

VCX Questionnaire

Please provide this information regarding your current environment and processing.

VCX Implementation Questionnaire

Please provide the following information regarding Visa Clearing Exchange (VCX) implementation to your Visa Representative.

Implementation Date Information:

Date of VCX Implementation (MM/DD/YYYY): ____ / ____ / ____

Implementation Request:

☐ Option 1	Migrate an existing CIB(s) from EP 4.0 to VCX
☐ Option 2	Migrate an existing CIB(s) from EP TC54 to VCX TC54
☐ Option 3	Set up a new CIB(s) on VCX (new BASE II endpoints)

BASE II CIB Information:

BASE II CIB	Client Name/BID	Region

Operating System Information:

Existing Edit Package Operating System: _____

VCX Operating System (Windows or Linux): _____

Version: _____

Environment:

VCMS: _____ Production: _____

Production Cutover Option:

☐ Option 1	TC 54 Records are Included in the UNDIFF File
☐ Option 2	Request VCX FFR Separately
☐ Option 3	TC 54 Records are delivered as a Custom File Type

NOTE: Reference VCX documentation for details as can be found on Visa Online (VOL).

Notices

User interfaces, menu options and steps depicted herein are illustrative only, may vary and are subject to change in Visa's sole discretion.

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